

# One shared structure to review

A practical working plan for connecting voice, AI extraction, backend, data and staff review.

**Purpose:** agree how our parts connect, what each owner will deliver next, and what must pass in the first joint demo. This document consolidates the discussion so far; it does not claim that every detail is approved or implemented.

## The product in one sentence

A patient speaks; AI prepares a draft intake form; staff see updates, correct the information and approve the record. AI does not make the final clinical triage decision.

## What we know - and what remains open

| Status                     | Current position                                                                                                                                          |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reported team direction    | Fazwan reports that the team has selected <b>FastAPI + SSE</b> . FastAPI is the backend framework; SSE carries live backend updates to the dashboard.     |
| Existing project baseline  | Five engineering roles, with Maryam covering demo/presentation. The earlier workflow includes follow-up questions, context retrieval and staff approval.  |
| Illustrative team example  | The shared <b>EXTRACTION_UPDATE</b> JSON shows the intended content. The teammate explicitly described it as an example; it is not yet a frozen contract. |
| Proposals in this document | Full current extraction state per update, version handling, evidence references, review protection and staged integration checks.                         |
| Still to confirm           | Database, final field/event names, authentication, hosting, coding support for Role 2, current readiness and the joint-test date.                         |

**SSE is one connection in the system.** It is not the database, the speech service or the AI model. Selecting SSE does not by itself remove Supabase or select another database.

**Reading guide:** pages 2-3 show the flow; page 4 assigns deliverables; pages 5-6 explain the data agreement; pages 7-8 cover the joint test and team confirmation; page 9 is Fazwan's Bahasa Melayu guide.

# The main intake flow

Target workflow. Role ownership is shown inside each box; live delivery and return actions are different paths.



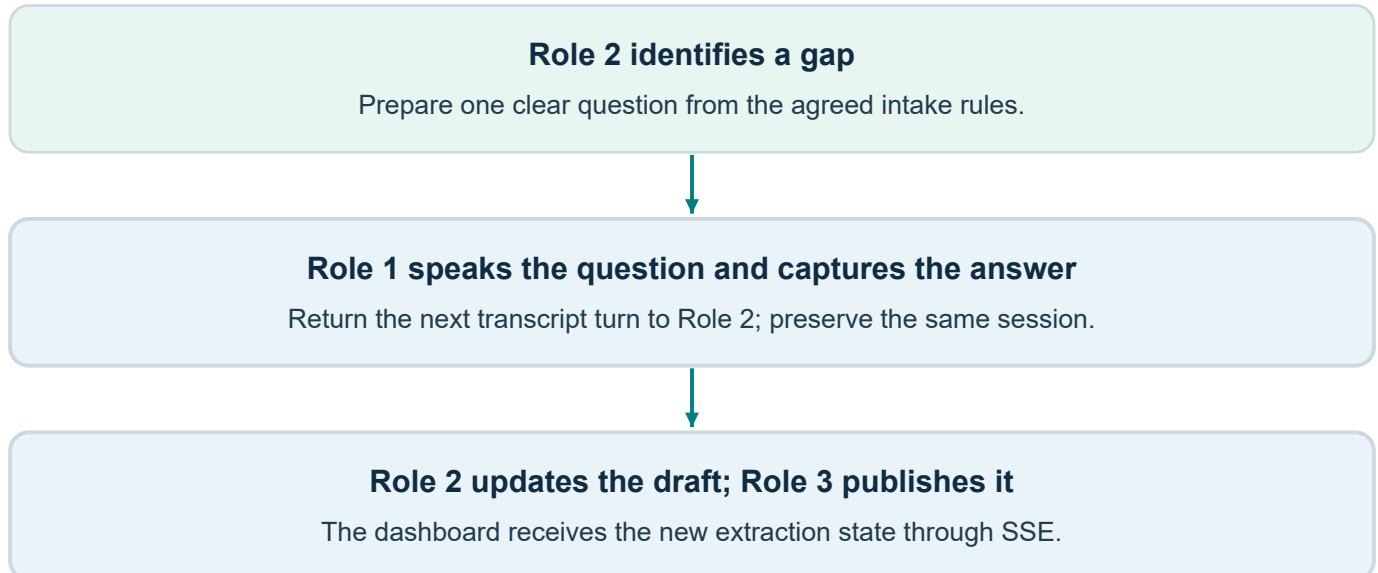
**Two important boundaries:** dashboard edits and approval return through an authenticated API, not through the one-way SSE stream. The dashboard does not write directly to the database.

This is a logical handoff diagram, not a prescribed network topology. SSE is highlighted only between backend and dashboard. The approved-record path does not prohibit saving drafts or events under a separately agreed retention policy.

# The two loops around the flow

Follow-up questions and record retrieval remain part of the proposed design. Neither is replaced by SSE.

## A. Ask for information that is missing



If the person cannot answer, keep the field unknown and surface the gap to staff. Do not invent an answer. Caregiver statements, staff observations and retrieved records must keep their own source labels.

## B. Retrieve context when authorised and available

| Step               | Owner and handoff                                                                                                                   |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 1. Request context | Role 2 requests an allowed lookup through Role 3. Matching a patient to a record needs an agreed identifier; do not guess identity. |
| 2. Execute lookup  | Role 3 checks permission and invokes Role 4's data interface. Tool status is recorded by the executing system.                      |
| 3. Return evidence | Role 4 returns a record with source metadata, or an explicit not-found/error result, through Role 3 to Role 2.                      |
| 4. Flag a conflict | Role 2 retains both statements if they disagree. Role 3 sends the updated draft to Role 5 for staff review.                         |

**Synthetic walkthrough:** transcript turn T1 says "no known allergies"; a deliberately seeded demo record R1 lists a Penicillin allergy. The draft shows both sources as an unresolved conflict. A lookup marked SUCCESS does not prove that either statement is clinically correct.

**Approval is a separate state.** Extraction can be in progress or ready for review; staff review can be pending, returned for more information or approved. Exact enum names are still open. A later AI update must not silently overwrite a staff correction or an approved record.

# Who delivers what next

Working ownership follows the repository roster. Each owner should confirm readiness, blockers and a test date.

| Owner                                                   | Deliverable and handoff                                                                                                                                                   | First evidence of readiness                                                                                                                      |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Role 1</b><br>Shahzaib Fraz<br>Voice / conversation  | Capture speech and return transcript turns with stable session/turn references to Role 2. Speak the response text supplied by Role 2.                                     | One synthetic spoken exchange becomes text; a follow-up question can be spoken and its answer captured.                                          |
| <b>Role 2</b><br>Fazwan Zainuddin<br>AI / agent         | Own intake fields, prompt/flow, extraction, source labels, gaps, conflict handling and expected outputs. Send drafts/tool requests to Role 3 and response text to Role 1. | Three transcript/output fixtures: sufficient information, missing information and conflicting sources. Run extraction against the agreed format. |
| <b>Role 3</b><br>Mozzam Shahid<br>Backend / integration | Own FastAPI, access checks, validation, tools, case versions, SSE delivery and staff-action APIs. Connect Roles 2, 4 and 5.                                               | Accept a synthetic draft, publish an update, receive a review action and return a confirmed save result.                                         |
| <b>Role 4</b><br>Ishaan<br>Data / persistence           | Confirm database and identifiers; provide authorised context lookup and approved-record storage. Define draft/event retention and audit storage with Role 3.              | Return one synthetic record and one not-found result; save an approved record once with its audit history.                                       |
| <b>Role 5</b><br>Jonathan<br>Dashboard / review         | Render draft updates and source/gap/conflict states. Send staff edits, information requests and approval to Role 3.                                                       | Display a sample, then a live update; show reconnection and save status; protect staff edits from newer AI drafts.                               |
| <b>Demo</b><br>Maryam Habib                             | Own the demo story, script, role transitions and presentation. Show human review clearly and label simulated components.                                                  | Rehearse one complete synthetic case. Prepare a transparent fallback if a live dependency fails.                                                 |

**Role 2 support must be explicit.** Fazwan can lead flow, requirements and expected-output review. The extraction runtime still needs implementation and testing. A coding partner has not been assigned in this discussion; the team should confirm one if needed.

**Working boundary:** Role 2 does not implement direct database access; Role 5 does not bypass the backend. Clinical approval permissions and mandatory approval fields need a designated domain reviewer. Names are from README.md; confirm any roster changes or Discord aliases.

# The team's example message

Reference sample supplied in chat on 31 August 2026. Formatting escapes removed; field names and values retained.

```
{
  "event_id": "evt_9876543210",
  "session_id": "sess_88f1a23b-4cde-4f56-9a1b-2c3d4e5f6a7b",
  "timestamp": "2026-08-31T12:00:00Z",
  "event_type": "EXTRACTION_UPDATE",
  "payload": {
    "status": "IN_PROGRESS",
    "patient_reported": {
      "chief_complaint":
        "Severe chest tightness and shortness of breath",
      "onset": "30 minutes ago",
      "severity": 8,
      "symptoms": [
        "chest pain",
        "dyspnea",
        "diaphoresis"
      ]
    },
    "missing_information": [
      "Known heart conditions",
      "Current medications"
    ],
    "conflicts_detected": [
      {
        "field": "allergies",
        "description":
          "Patient reported no allergies, but record shows Penicillin allergy."
      }
    ],
    "tool_executions": [
      {
        "tool_name": "retrieve_patient_context",
        "status": "SUCCESS",
        "executed_at": "2026-08-31T11:59:58Z"
      }
    ]
  }
}
```

**Read it as:** the outer fields identify the update and session. The payload contains the draft information, gaps, conflicts and actual tool activity for the dashboard.

**Example, not proof:** this sample does not demonstrate that extraction or retrieval has run. Do not treat its SUCCESS value as a real execution result. Names/addresses were suggested for later addition, but their necessity, source and handling are not agreed.

The event wrapper and execution metadata are system-owned. Patient-reported facts come from source text; conflicts require both supporting sources. This JSON is message content, not the complete SSE wire format.

# Agree the connection contract

A contract is the shared format and meaning of the data. These changes are proposals for the owners to review.

| Decision                | Proposed MVP rule / owner                                                                                                                                                                                                                        |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| One format              | Roles 2, 3 and 5 agree one schema. The team sample uses snake_case and EXTRACTION_UPDATE; the existing draft uses camelCase and case.updated. Map deliberately; do not assume the two objects are interchangeable.                               |
| Update meaning          | Use the full current extraction state per update for the first demo. Replace only the extraction view, not staff edits or approval state. If partial updates are preferred, define merge and deletion rules explicitly.                          |
| Identity and ordering   | Role 3 owns event identity and authoritative ordering. Agree session-to-case mapping and a case_version field. Ignore duplicate or older updates; define a snapshot/replay path after reconnection.                                              |
| Unknown and evidence    | Role 2 must not fill gaps by guessing. Agree null/unknown handling and source references for each relevant field. Preserve patient, caregiver, staff and retrieved-record origins; retain source/confidence information from the existing draft. |
| Field and state meaning | Define severity scale and associated symptom, onset reference time, required fields, and extraction vs review states. Agree separate error/not-found outcomes; extraction completion must not imply staff approval.                              |
| Review and save         | Roles 3 and 5 agree edit paths and version checks. Block stale approval; protect confirmed edits from AI overwrite. Save once, keep an audit trail and show success only after the backend confirms storage.                                     |
| Access and hosting      | Roles 3 and 5 agree stream authentication. Native browser EventSource cannot attach arbitrary bearer headers; choose suitable secure cookies or an authorised streaming client. Test proxy buffering, reconnection and HTTPS on the chosen host. |
| Storage and privacy     | Role 4 confirms the database; SSE does not settle this choice. Agree draft/event retention, synthetic identity fields and record matching. No real patient content or tokens in repo, demo fixtures or ordinary logs.                            |

## How the existing draft fits

Keep the useful draft concepts: sources, confidence, information gaps, conflicts, human review and version protection. Agree how they map into the teammate's message before editing the schemas. This PDF does not modify or replace the existing JSON contracts.

**Backend-owned interfaces to confirm:** submit a draft; request context; receive live events; submit staff review; obtain the latest state after reconnect. Existing route names remain proposals. Developers should confirm the FastAPI version and SSE implementation supported by their environment.

# Build toward one joint demo

Use small integration checkpoints. No date or completion claim is implied; owners supply readiness estimates.

| Checkpoint                                              | Work to connect                                                                                         | Evidence before moving on                                                                                     |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>1. Shared fixture</b><br>Roles 2 / 3 / 5             | Agree one sample case, field meanings and initial status. Role 4 provides any synthetic lookup fixture. | The same expected data is understood by extraction, backend and dashboard owners.                             |
| <b>2. Live delivery</b><br>Roles 3 / 5                  | Backend sends the agreed sample through SSE while extraction is still being built.                      | Dashboard shows the sample and one changed value without manual refresh. Any simulated component is labelled. |
| <b>3. Voice and AI</b><br>Roles 1 / 2 / 3               | Feed transcript turns into extraction, publish results, ask one follow-up and process its answer.       | Information updates from actual transcript input; gaps shrink appropriately; unsupported facts do not appear. |
| <b>4. Context and review</b><br>Roles 2 / 3 / 4 / 5     | Use a seeded record to produce a source-backed conflict; staff review and correct it.                   | Both sources remain visible; staff action and reviewer identity are recorded; AI does not approve.            |
| <b>5. Save and rehearse</b><br>Roles 3 / 4 / 5 + Maryam | Approve the case, store it and rehearse the complete story.                                             | One saved record and audit history; dashboard shows confirmed success; the demo can be repeated.              |

## Joint-test checklist

- Complete, missing-information and conflicting-source fixtures produce the agreed outputs.
- Disconnect/reconnect preserves the latest draft; duplicate or older events do not roll it back.
- A stale staff edit is flagged; new AI data does not erase a staff correction or approved record.
- Lookup failure/not-found is visible and is never presented as a successful patient match.
- Unauthorised case access is rejected; no patient content or tokens appear in ordinary logs.
- Failed storage remains visibly unsaved; retries do not create duplicate approved records.

**Demo target:** one clearly labelled synthetic case moves from spoken intake to a live draft, one follow-up, human review and a confirmed saved record. This is a prototype integration check, not a claim of clinical readiness.

# Team review and confirmation

Please review the structure, correct anything that differs from your implementation, and report what is ready.

| Please confirm                                                                               | Who responds                                  |
|----------------------------------------------------------------------------------------------|-----------------------------------------------|
| Do we accept the main flow, follow-up loop and authorised context loop on pages 2-3?         | All owners                                    |
| Are the roster and boundaries correct? Who will pair on Role 2 implementation if needed?     | All owners; coding support still unassigned   |
| Which schema/event names, update meaning, identifiers, source fields and states will we use? | Roles 2 / 3 / 5; Role 4 for identifiers       |
| Which database, access method, retention policy, host and stream authentication will we use? | Roles 3 / 4 / 5                               |
| Who validates the approval fields, triage vocabulary and authorised reviewer roles?          | Team-designated domain reviewer               |
| What is ready now, what is blocked, and when can we run checkpoint 2 and the full demo?      | Each owner; Maryam coordinates demo readiness |

## Suggested reply format

**My role:** ...  
**Structure:** agree / agree with changes listed below  
**Ready now:** ...  
**Next deliverable and target date:** ...  
**Dependency or help needed:** ...  
**Change requested, with page reference:** ...

## Sources and scope

Project sources: README.md (role roster); contracts/ROLE2\_ROLE3\_ROLE5\_HANDOFF.md and both JSON schemas (draft interfaces); AI\_Emergency\_Intake\_Team\_Workflow\_Roles.pdf dated 30 Aug 2026 (earlier proposal); teammate example and Fazwan's update shared in this conversation on 31 Aug 2026.

Technical references, checked 31 Aug 2026:

1. FastAPI SSE: [fastapi.tiangolo.com/tutorial/server-sent-events/](https://fastapi.tiangolo.com/tutorial/server-sent-events/)
2. MDN SSE: [MDN - Using server-sent events](#)
3. MDN EventSource constructor: [MDN - EventSource constructor and options](#)
4. Supabase: [supabase.com/docs](https://supabase.com/docs)

**Status control:** FastAPI + SSE is recorded as the direction reported by Fazwan, not independent evidence of every teammate's approval. Other recommendations remain open until owners confirm. No backend, database, deployment or live test was audited for this document; no runtime or schema was changed.

# Panduan ringkas untuk Fazwan

Bahasa Melayu | Apa maksud keputusan ini dan bagaimana membawa perbincangan team ke langkah seterusnya.

**Bayangkan borang yang terisi semasa pesakit bercakap.** Shahzaib mengurus suara. Bahagian AI anda menyusun maklumat. Backend Mozzam menghantar update melalui SSE. Dashboard Jonathan memaparkannya. Staf menyemak dan mengesahkan; backend menghantar rekod untuk disimpan oleh bahagian Ishaan.

## Apa yang berubah dengan SSE?

SSE ialah cara update bergerak dari backend ke dashboard. Ia tidak menentukan database dan tidak mengambil alih kerja AI atau suara. Jika yang anda maksudkan sebelum ini ialah Supabase, kita belum boleh kata ia dibuang hanya kerana memilih SSE.

## Tiga tindakan anda sekarang

- **1. Kongsi PDF ini untuk semakan.** Minta team sahkan struktur dan beri pembetulan. Elakkan menyatakan semua butiran sudah dipersetujui; beberapa keputusan masih terbuka pada halaman 8.
- **2. Sediakan contoh hasil yang anda mahu.** Untuk setiap percakapan rekaan, tulis maklumat yang patut masuk ke borang, apa yang belum diketahui dan apa yang bercanggah. Ini menjadi panduan untuk membina serta menguji AI.
- **3. Pastikan kerja dan bantuan jelas.** Minta setiap owner beritahu apa yang siap, apa yang belum dan tarikh sasaran ujian. Jika anda memerlukan bantuan coding extraction, nyatakan terus supaya kerja itu mempunyai pemilik.

## Contoh mudah untuk mula

**Pesakit rekaan:** "Dada saya terasa ketat sejak 30 minit lalu."

**Hasil dijangka:** aduan = dada ketat; mula = 30 minit lalu.

**Belum diketahui:** maklumat lain yang belum disebut kekal kosong/unknown mengikut format yang dipersetujui.

**Peraturan:** jangan tambah simptom, alergi atau keputusan triage yang tidak disokong sumber.

## Mesej untuk dihantar bersama PDF

Hi team, I have consolidated our discussions so far into this updated workflow PDF, reflecting FastAPI + SSE as the current direction. It includes the main flow, follow-up and context-retrieval loops, role handoffs, proposed data rules and the first integration test.

Could everyone review it and confirm whether we can follow this structure, or list the changes needed? Please also share what is ready, your next deliverable, blockers and a target test date. The database and final data contract are still marked for confirmation. I will focus on the intake flow, expected extraction outputs and test cases; please confirm who can pair with me on extraction implementation if needed.